

Retrieval-Augmented Generation (RAG) Guide

Chapter 1: What is RAG?

Retrieval-Augmented Generation (RAG) is an AI architecture that enhances Large Language Models by fetching relevant factual chunks from external knowledge bases before generating an answer. This significantly reduces LLM hallucinations and allows models to access private or domain-specific documents.

Chapter 2: Text Processing & Chunking Strategies

Document ingestion requires splitting large texts into manageable chunks.

- **Chunk Size:** Typically between 500 to 1000 characters.
- **Chunk Overlap:** Typically 100 to 200 characters to prevent loss of semantic context across boundaries.
- **Recursive Chunking:** Splits text at paragraphs, sentences, and words recursively.

Vector Stores & Retrieval Accuracy

Chapter 3: Vector Databases (FAISS & Chroma)

Embeddings convert text chunks into high-dimensional numerical vectors. Vector databases store these vectors and perform fast Approximate Nearest Neighbor (ANN) search using distance metrics like Cosine Similarity or Euclidean Distance.

- **FAISS** (Facebook AI Similarity Search): Extremely fast GPU/CPU vector indexing library.
- **ChromaDB**: Open-source developer-friendly vector database with persistence capabilities.

Chapter 4: Evaluation Metrics

Key metrics for evaluating RAG systems include Context Precision, Context Recall, Answer Faithfulness, and Answer Relevance. Strict prompt templates ensure the LLM returns 'I couldn't find this information' when context is missing.